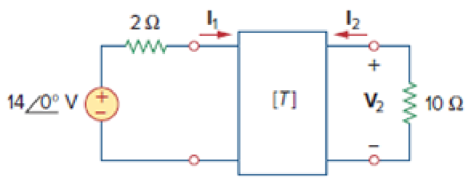
Chapter 19, Problem 9PP

Problem

Find **I1** and **I2** if the transmission parameters for the two-port in Fig. 19.36 are

$$\begin{bmatrix} 5 & 10\,\Omega \\ 0.4\,\text{S} & 1 \end{bmatrix}$$

Figure 19.36



Step-by-step solution

Step 1 of 6

Substituting the given ABCD parameters into

$V_1 = AV_2 - BI_2$

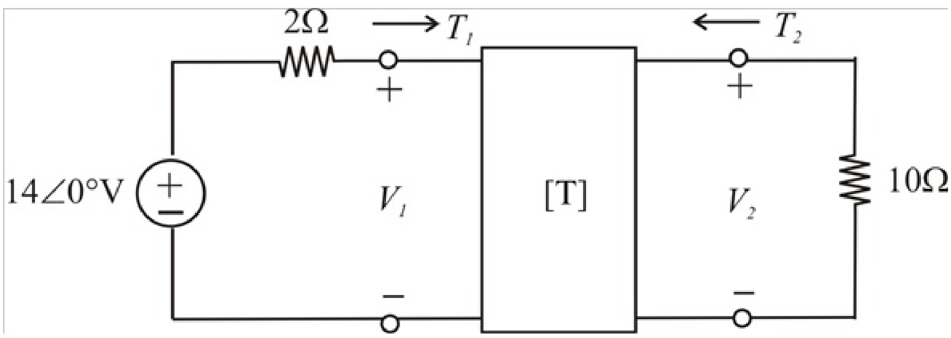
$I_1 = CV_2 - DI_2$

$V_1 = 5V_2 - 10I_2 \quad \rightarrow (1)$

$I_1 = 0.4V_2 - I_2 \quad \rightarrow (2)$

Step 2 of 6

Given circuit is



Step 3 of 6

Apply KVL, we get

$-14 + 2I_1 + V_1 = 0$

$V_1 = 14 - 2I_1$

Then

$V_2 = -10I_2$

Step 4 of 6

Substituting V_1, V_2 in equation (1) and (2), then $14-2I_1=5V_2-10I_2$

$$14-2I_1=5(-10I_2)-10I_2$$

$$14-2I_1+60I_2=0$$

$$-I_1+30I_2=-7$$

$$I_1-30I_2=7 \qquad \rightarrow(3)$$

Step 5 of 6

Hence

$$I_1=0.4(-10I_2)-I_2$$

$$I_1+5I_2=0 \qquad \rightarrow(4)$$

By solving (3) and (4) we obtain

$$\begin{array}{r} I_1-30I_2=7 \\ \underline{I_1+5I_2=0} \\ -35I_2=7 \end{array}$$

$$I_2=-\frac{1}{5}$$

$$\therefore \boxed{I_2=-0.2\text{A}}$$

Step 6 of 6

From equation (4), we get

$$I_1+5(-0.2)=0$$

$$\therefore \boxed{I_1=1\text{A}}$$